EXPERIMENT NO: 7

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Subject Name: System Design
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Aim

To design a scalable ride-sharing platform similar to Uber/Ola that enables real-time ride booking, driver matching, fare estimation, and live tracking with high availability and low latency.

Objectives

- Understand the architecture of real-time ride-sharing systems.
- Identify functional requirements such as ride booking and driver matching.
- Identify non-functional requirements including scalability, latency, and availability.
- Analyze consistency trade-offs using CAP theorem.
- Design REST APIs for rider and driver workflows.

Procedure

1. Studied real-world ride-sharing platforms such as Uber and Ola.
2. Identified core entities such as Rider, Driver, Ride, Location, and Fare.
3. Analyzed real-time location tracking using WebSockets.
4. Collected functional and non-functional requirements.
5. Designed APIs for ride booking, matching, and tracking.
6. Evaluated proximity-based driver matching algorithms.
7. Analyzed consistency constraints in distributed ride allocation.

Functional Requirements

1. Rider should be able to get fare estimation based on pickup and drop location.
2. Rider should be able to request a ride.
3. Rider should be able to choose ride categories (car, bike, auto, etc).
4. Driver should be able to accept or reject ride requests.
5. System should match riders with nearby available drivers.
6. System should support real-time tracking of rides.
7. Riders and drivers should be able to rate each other after trip completion.
8. System should support multiple payment methods.

Core Entities of the System

- User / Rider
- Driver

- Location (Pickup & Drop-off)
- Fare
- Ride / Trip

API Design

1. Rider API

A. Get Fare Estimation:
`GET /api/v1/fare?pickupLat={}&pickupLng={}&dropLat={}&dropLng={}`

B. Request Ride: `POST /api/v1/rides/request`

C. Ride History: `GET /api/v1/rides/history`

D. Cancel Ride: `POST /api/v1/rides/{ride_id}/cancel`

E. Rate Ride: `POST /api/v1/rides/{ride_id}/rating`

2. Driver API

A. Update Location (WebSocket): `WS /drivers/location`

B. Accept/Reject Ride: `POST /api/v1/rides/{request_id}/respond`

C. Start Ride: `POST /api/v1/rides/{ride_id}/start`

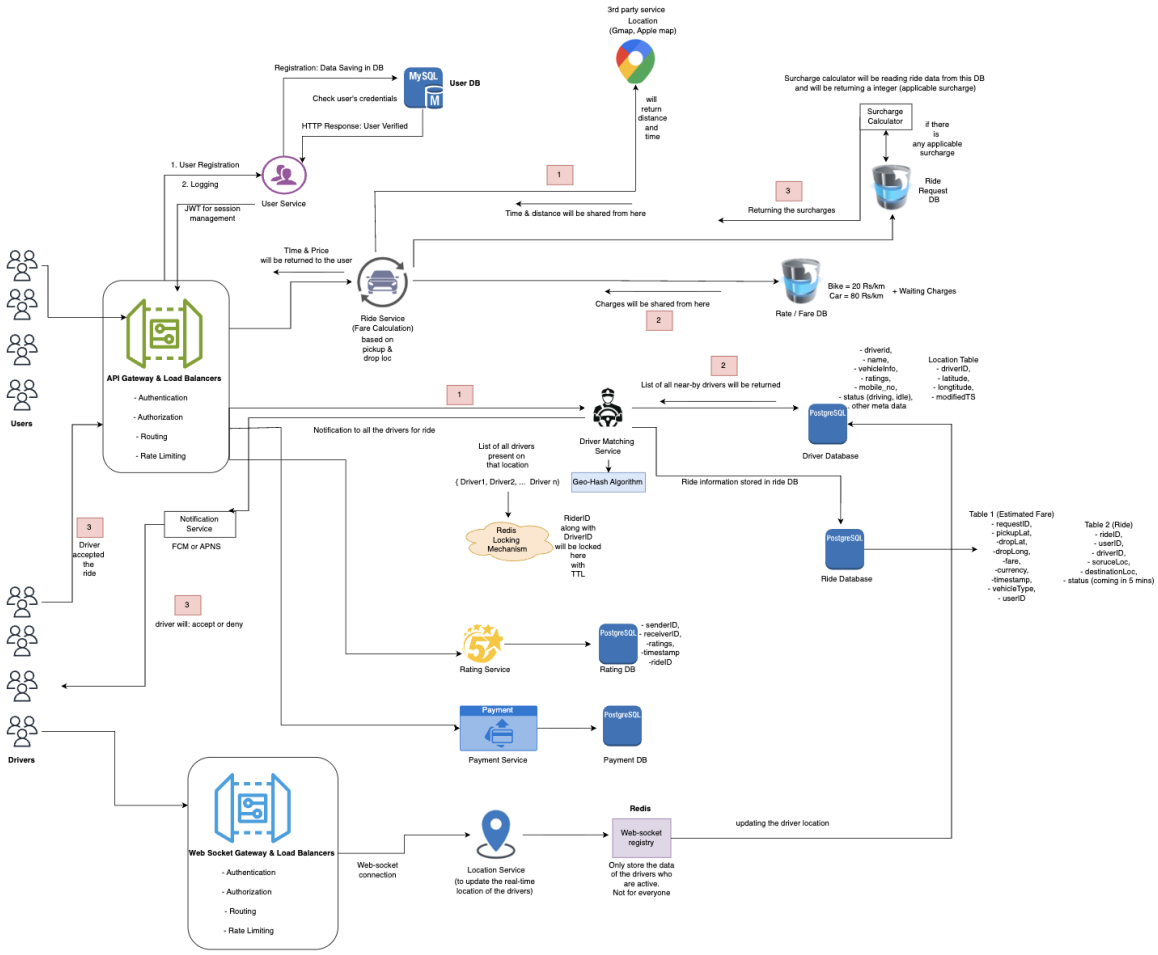
D. End Ride: `POST /api/v1/rides/{ride_id}/end`

Non-Functional Requirements

1. System should support up to 1.5 million users and drivers.
2. High availability for riders (eventual consistency).
3. Strong consistency for driver assignment (no double booking).
4. Driver matching latency should be less than 1 minute.
5. Real-time location updates should have minimal delay.

High Level Design (HLD)

The system consists of client applications (rider and driver apps), an API Gateway, and microservices such as User Service, Ride Service, Matching Service, Location Service, and Payment Service. Driver locations are updated via WebSockets and stored in a geo-indexed datastore. The Matching Service finds nearby drivers and assigns rides. Asynchronous processing using message queues ensures scalability, while payments and ratings are handled after trip completion.



Outcome

- Designed a scalable ride-sharing system with real-time capabilities.
- Understood trade-offs between consistency and availability.
- Implemented structured API design for rider and driver workflows.
- Learned about real-time systems using WebSockets and geo-indexing.